



Kolhapur Institute of Technology's

College of Engineering (Autonomous), Kolhapur

**“Smart Campus Canteen Management & Food Request
Routing System ”**

(Initial-Report)

Mini Project-III (Android)

(UAMIL0571)

By

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1. Abstract

The **Smart Campus Canteen Management & Food Request Routing System** is a centralized web and mobile application developed to streamline food request management within a college campus. The proposed system addresses the limitations of conventional manual ordering methods by integrating multiple campus canteens, food courts, and hostel messes into a single digital platform. Unlike commercial food delivery applications, the system follows a centralized request-routing model in which food requests are first transmitted to a central college server. The server authenticates users, verifies canteen availability, validates hostel delivery eligibility where applicable, and forwards the request to the selected canteen for processing. Based on current operational conditions, the canteen may accept or reject the request and communicate the estimated preparation time, which is then relayed to the user through the central server. Upon acceptance, the request is fulfilled through either pickup or hostel delivery.

The proposed system is designed to serve students, hostel residents, faculty members, staff, canteen operators, and college administrators through a unified platform. It incorporates features such as digital menu management, real-time request tracking, QR code-based pickup verification, secure online payment integration, order history, feedback and rating mechanisms, and an administrative dashboard for monitoring operations and generating analytical reports. By centralizing communication between users and campus food outlets, the system aims to reduce waiting time, improve operational efficiency, minimize congestion during peak hours, and enhance the overall food service experience within the institution.

2. Introduction

The increasing adoption of digital technologies has transformed service management across educational institutions, including academic administration, library systems, attendance management, and fee payment. However, food service operations in many colleges continue to rely on conventional manual processes, resulting in inefficiencies for both users and canteen operators. Institutions with multiple food courts, departmental canteens, and hostel messes often operate these facilities independently, without a centralized platform for communication, request management, or service monitoring.

In the existing system, students, faculty members, and staff are required to physically visit individual canteens to check menu availability, place requests, and wait in queues without any prior information regarding preparation time or crowd levels. Hostel residents face additional inconvenience due to the absence of a structured mechanism for requesting food delivery to their hostels. Consequently, users experience unnecessary waiting, inconsistent service quality, and limited convenience, particularly during peak hours.

Similarly, canteen operators lack a unified digital platform to manage incoming requests, update menu availability, indicate stock shortages, or regulate request intake during periods of high demand. From an administrative perspective, the absence of centralized monitoring makes it difficult to evaluate canteen performance, analyze service trends, monitor operational efficiency, or generate meaningful reports for decision-making.

To address these challenges, this project proposes the **Smart Campus Canteen Management and Food Request Routing System**, a centralized web and mobile application developed exclusively for use within the college campus. Unlike commercial food delivery platforms that facilitate direct transactions between customers and restaurants, the proposed system follows a centralized request-routing architecture. Every food request is first transmitted to a central college server, where it is authenticated, validated, and forwarded to the selected canteen for processing. The canteen then responds with the request status and estimated preparation time, after which the request is fulfilled through either pickup or hostel delivery, depending on the user's preference and service availability.

By integrating multiple campus food outlets into a single digital platform, the proposed system aims to improve operational efficiency, reduce waiting time, enhance communication between users and canteen operators, and provide the college administration with a comprehensive platform for monitoring, management, and data-driven decision-making. This report presents the analysis, design, and proposed implementation of the system, highlighting its objectives, architecture, functional components, and anticipated benefits to all stakeholders.

3. Problem Statement

The existing food service system within the college campus operates through multiple independent canteens, food courts, and hostel messes, each functioning without a centralized digital platform. As a result, the ordering and service process is largely manual, leading to operational inefficiencies and inconvenience for both users and service providers.

Students, faculty members, and staff are required to visit individual canteens to check menu availability, place food requests, and wait in queues without prior knowledge of preparation time or service availability. This often results in increased waiting time, overcrowding during peak hours, and an inconsistent user experience. Hostel residents face additional challenges due to the absence of a structured mechanism for requesting food delivery to hostel premises.

From the operational perspective, canteen operators lack an integrated system to manage incoming requests, update menu availability, indicate stock shortages, regulate request intake during busy periods, or communicate estimated preparation times effectively.

Furthermore, the college administration does not have a centralized mechanism to monitor the performance of multiple food outlets, analyze operational data, or make informed decisions regarding service quality, resource allocation, and overall canteen management.

Therefore, there is a need for a centralized campus food request management system that can integrate all campus food outlets, streamline communication between users and canteen operators, improve operational efficiency, reduce waiting time, and provide administrators with comprehensive monitoring and analytical capabilities.

4. Existing System

The existing food service system within the college campus primarily operates through conventional manual procedures. Each canteen, food court, or hostel mess functions independently, with users required to visit the respective outlet to view the menu, place food requests, and collect their orders. During peak hours, some canteens may implement token-based or queue-based service mechanisms; however, these systems remain isolated and are not integrated across the campus.

The absence of a centralized digital platform makes it difficult for students, faculty members, and staff to access information such as menu availability, estimated preparation time, or operational status before visiting a canteen. Similarly, hostel residents have limited or no access to a formal food delivery mechanism, resulting in inconvenience during academic hours or late-night periods.

From the service provider's perspective, canteen operators rely on manual order management, making it difficult to regulate incoming requests, update menu availability, communicate preparation delays, or manage customer flow efficiently. Furthermore, the college administration lacks a centralized system to monitor canteen operations, analyze service performance, or generate reports for effective decision-making.

Consequently, the existing approach leads to longer waiting times, overcrowding during peak hours, inefficient communication, and limited operational transparency.

Limitations of the Existing System

1. No centralized platform integrating all campus canteens and food courts.
2. No real-time information regarding menu availability or preparation time.
3. No structured mechanism for hostel food delivery.
4. Manual request handling leading to increased waiting time and service delays.
5. Lack of authenticated user identification linked to institutional credentials.
6. No centralized monitoring or analytical dashboard for college administration.
7. Limited or no support for digital payment integration.

8. Absence of an organized feedback and grievance management system.
9. Inefficient communication between users and canteen operators.
10. No historical data for analyzing demand patterns or service performance.

Table 1: Comparison of Systems

Aspect	Existing Manual System	Generic Food Delivery Apps (Swiggy/Zomato)	Proposed System
Authentication	None	Phone/email based	College ID linked
Order Model	Walk-in / informal	Direct instant order	Request → Server → Canteen approval
Hostel Delivery	Not supported	Not applicable (No campus service integration)	Built-in eligibility-checked delivery
Admin Monitoring	None	None (No institutional integration)	Centralized dashboard
Vendor Scope	Campus canteens (unconnected)	External restaurants (No campus canteen/mess integration)	Internal campus canteens, food courts, and messes
Cost to Institution	None, but inefficient	Not applicable	One-time + maintenance cost, campus-owned
Service Availability	Limited (Day only)	Limited (upto hostel policy hours)	24/7 (Day & Night Canteens)

5. Proposed System

The proposed system introduces a centralized request-routing architecture in which students, hostel residents, faculty, and staff submit food requests through a single application. Every request passes through a central college server responsible for user authentication, canteen availability verification, and, where applicable, hostel delivery eligibility validation, before being forwarded to the selected canteen. The canteen operator can accept or reject the request, modify the estimated preparation time, mark specific items as unavailable, or temporarily pause incoming requests. Once accepted, the system communicates the estimated preparation time to the requester and sends a notification when the request is ready. The requester can then either collect the food using QR code-based verification or receive it through hostel delivery, where available.

This **request-approval model**, rather than an **instant-order model**, is the fundamental architectural distinction between the proposed system and commercial food delivery platforms. It is specifically designed to reflect the operational requirements of campus canteens, where food preparation depends on available resources, current workload, and service capacity rather than guaranteed order fulfillment.

5.1 Objectives

The primary objectives of the proposed system are:

1. To centralize food request management across all campus canteens, food courts, and hostel messes through a single application.
2. To reduce physical queuing time by enabling advance food requests with real-time status updates.
3. To introduce a structured and eligibility-based hostel delivery mechanism.
4. To provide canteen operators with digital tools for managing food requests, menu availability, inventory status, and operational capacity.
5. To equip the college administration with centralized monitoring, reporting, and analytical capabilities across all campus food service outlets.
6. To improve transparency, accountability, and user satisfaction through request history, feedback, and rating mechanisms.

6. Scope

The scope of the proposed system defines the functionalities that will be implemented in the current phase of the project, as well as the features that are intentionally excluded.

- In Scope

1. Centralized food request routing across multiple campus canteens and food courts.
2. Integration of hostel messes with eligibility-based hostel delivery.
3. Digital menu management by canteen operators.
4. Real-time notifications for request status and estimated preparation time.
5. QR code-based verification for food pickup.
6. Secure in-app payment processing.
7. Feedback, ratings, and request history management.
8. Administrative dashboard for monitoring, reporting, and analytics.

- Out of Scope

1. Integration with off-campus restaurants or third-party food delivery platforms.
2. Food delivery outside the college campus.

3. Dynamic or surge pricing mechanisms.
4. Multi-campus deployment (limited to a single-campus implementation in the current phase).

6.1 Stakeholders

Stakeholder	Role in System
Student	Submits food requests, tracks request status, makes digital payments, views request history, and provides feedback and ratings.
Hostel Resident	Performs all student functions, with additional access to hostel delivery services and hostel address management.
Faculty	Submits food requests, primarily for pickup, and may receive priority service where applicable.
Staff	Submits food requests and accesses the system similarly to faculty members.
Canteen Operator	Manages digital menus, accepts or rejects incoming requests, updates estimated preparation time, manages item availability, and controls request intake.
Administrator	Monitors all campus canteens, manages users and canteens, views analytics and reports, and handles administrative oversight and escalations.

6.2 Assumptions

The proposed system is developed based on the following assumptions:

1. All users possess valid college-issued credentials for authentication and system access.
2. Each canteen has at least one operator-side device (computer, tablet, or smartphone) to manage incoming food requests.
3. The campus has reliable internet connectivity to support real-time request processing and notifications.
4. Canteen operators regularly update menu availability and request status.
5. Users have access to smartphones or web-enabled devices to use the application.

6.3 Constraints

The development and implementation of the proposed system are subject to the following constraints:

1. The initial deployment is limited to a single college campus.
2. Successful operation depends on the willingness of canteen operators to adopt the digital workflow.
3. Payment integration is restricted to gateways approved by the institution.
4. The system's performance depends on the availability of stable internet connectivity within the campus.
5. Hostel delivery services are subject to institutional policies and operational feasibility.

This version reads more like a software engineering report while remaining concise enough for an initial synopsis.

7. Stages of the Project (Project Methodology)

The project follows a structured lifecycle, aligned with standard software engineering practices and adapted to an academic final-year project timeline:

1. **Requirement Gathering & Analysis** – Conducting informal surveys and interviews with students, canteen staff, and college administration to identify functional and non-functional requirements while reviewing the limitations of the existing manual process.
2. **Feasibility Study** – Assessing the technical, operational, and economic feasibility of the proposed system before committing to the design phase.
3. **System Design** – Producing the system architecture, database schema, and UI/UX wireframes, and developing the system architecture, use case diagrams, and data flow diagrams presented in this report.
4. **Technology Selection** – Finalizing the frontend, backend, database, real-time notification mechanism, and payment gateway to be used.
5. **Implementation (Development)** – Module-wise development covering the Student Module, Hostel Module, Canteen Operator Module, and Administrator Module.
6. **Testing** – Conducting unit testing of individual modules, integration testing across the request-routing pipeline, and user acceptance testing with a pilot canteen.
7. **Deployment** – Rolling out the system to one canteen and one hostel mess as a pilot before expanding it campus-wide.

8. **Maintenance & Feedback Iteration** – Ongoing monitoring after deployment, with refinements driven by real user feedback and analytics.

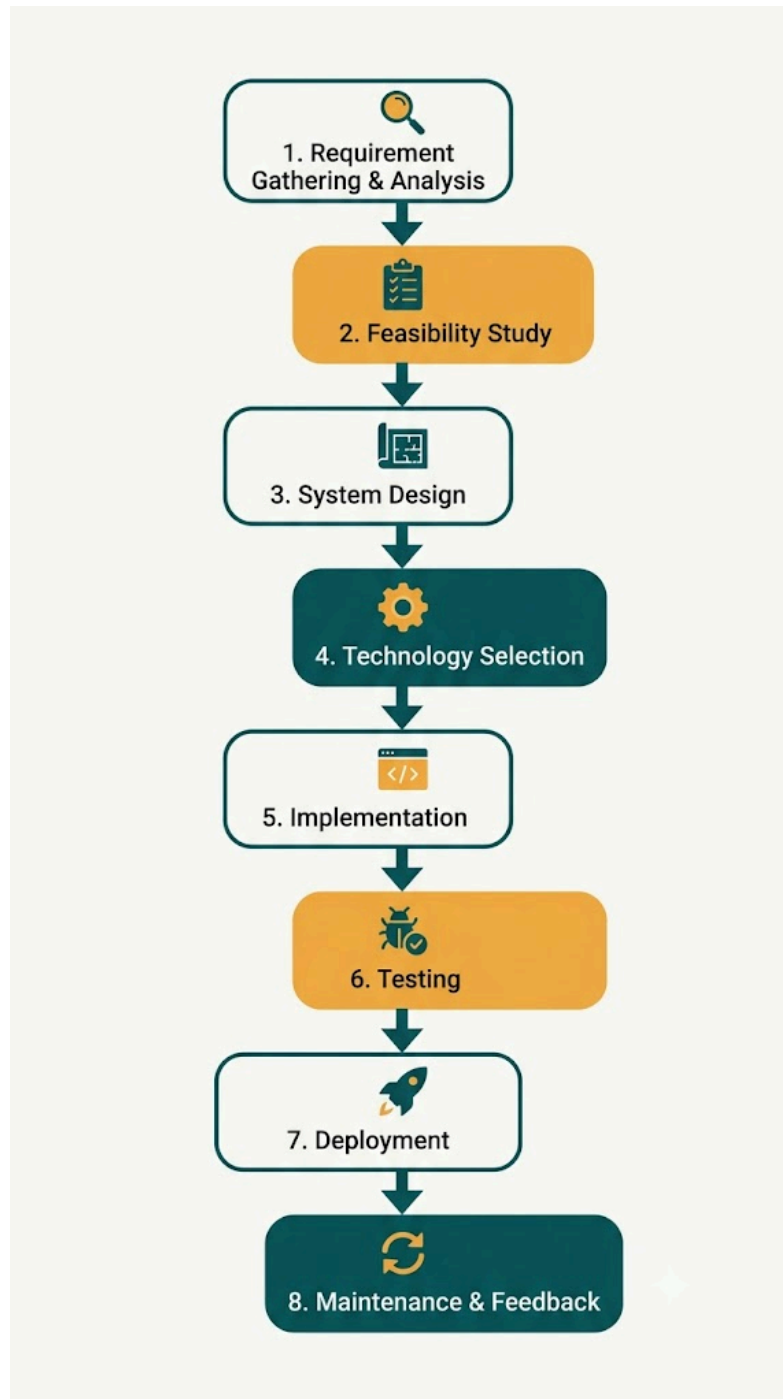


Fig 1.: Project Stages Timeline

8. System Architecture

The system follows a layered client-server architecture. At the client layer, three distinct interfaces exist: the Student/Hostel/Faculty/Staff-facing mobile or web application, the

Canteen Operator dashboard, and the Administrator dashboard. All client requests pass through an API layer into the Central Server, which houses the core business logic: authentication and role-based access control, request validation, hostel eligibility verification, and routing logic that directs each request to the appropriate canteen. The Central Server interacts with a database layer storing user records, canteen and menu data, request/order records, and feedback data. Two external-facing services support the core logic: a notification service (for push/SMS/in-app alerts at each stage of the request lifecycle) and a payment gateway (for processing in-app payments securely).

This layered separation ensures that no client ever communicates directly with a canteen; every interaction is mediated by the Central Server, which is the defining architectural principle of this system and the basis for its centralized monitoring and control capabilities.

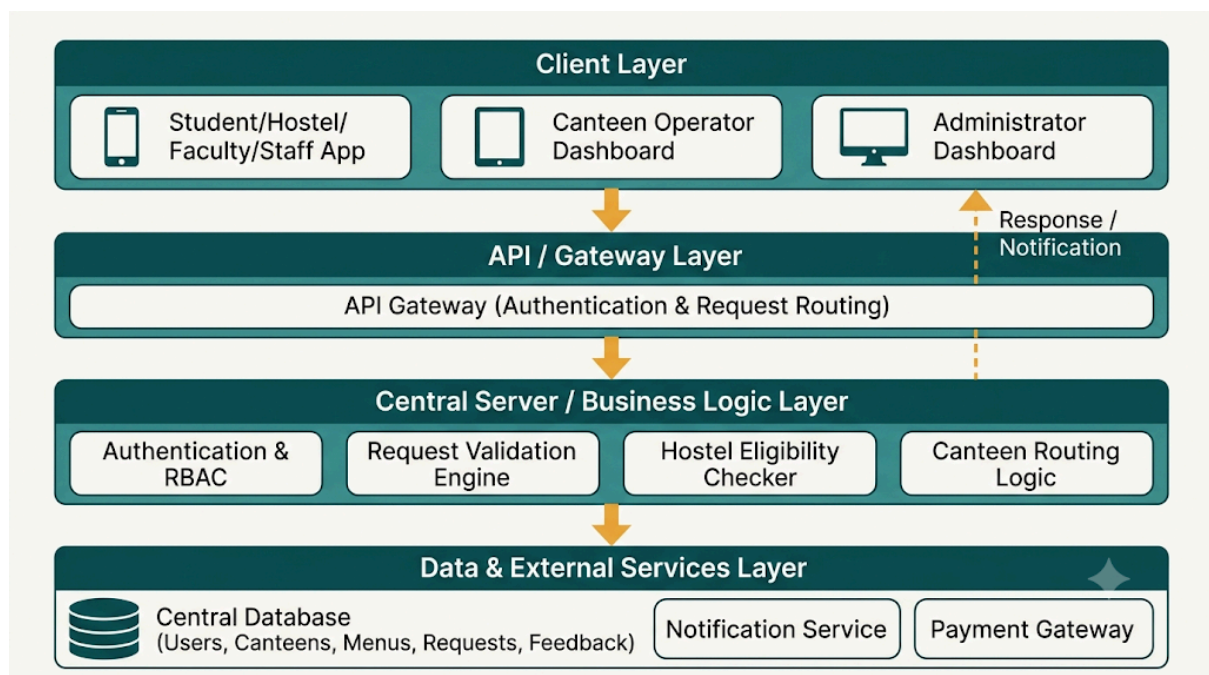


Fig 2.: System Architecture

9. Use Case Diagram

The system supports six actors, each with distinct interactions. Students and hostel residents can log in, browse digital menus across canteens, submit a food request, track its status in real time, make in-app payments, and provide feedback after fulfillment; hostel residents additionally have their delivery eligibility checked as an extension of the request submission use case. Faculty and staff follow a similar but typically pickup-only flow. Canteen operators can view incoming requests, accept or reject them, update estimated

preparation times, mark specific menu items as unavailable, and temporarily pause new incoming requests during high-demand periods. Administrators can monitor all canteens in real time, view cross-canteen analytics and reports, manage user accounts and permissions, and handle escalated complaints or disputes.

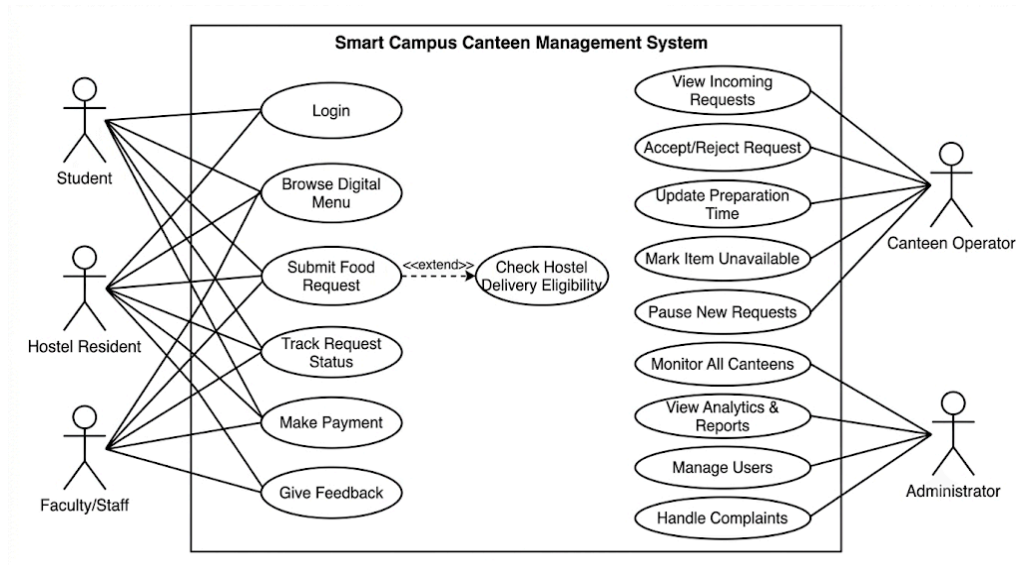


Fig 3.: Use Case Diagram

10. Data Flow Diagram

10.1 Level 0 (Context Diagram)

At the highest level of abstraction, the entire system is represented as a single process — "Smart Campus Canteen Management System" — surrounded by four external entities: Student/Hostel Resident/Faculty/Staff (submitting requests and receiving responses), Canteen Operator (receiving requests and sending decisions), Administrator (receiving reports and sending management commands), and Payment Gateway (processing transactions).

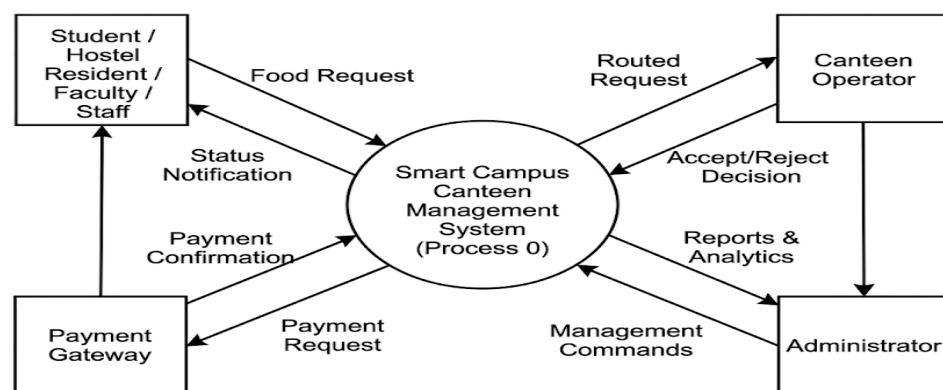


Fig 4.: Data Flow Level-0 (Context Diagram)

10.2 Level 1 (Decomposed Process View)

At Level 1, the single system process is decomposed into five core sub-processes:

- (1) Authentication & Authorization, which validates user credentials against the User data store;
- (2) Request Validation & Routing, which checks canteen availability and hostel eligibility before forwarding requests to the appropriate canteen;
- (3) Canteen Response Handling, which processes accept/reject/modify decisions from canteen operators and updates the Request data store;
- (4) Notification Dispatch, which triggers alerts to users at each stage of the request lifecycle; and
- (5) Payment Processing, which handles transaction initiation and confirmation with the external Payment Gateway. Three data stores support these processes: the User Database, the Canteen & Menu Database, and the Request/Order Database.

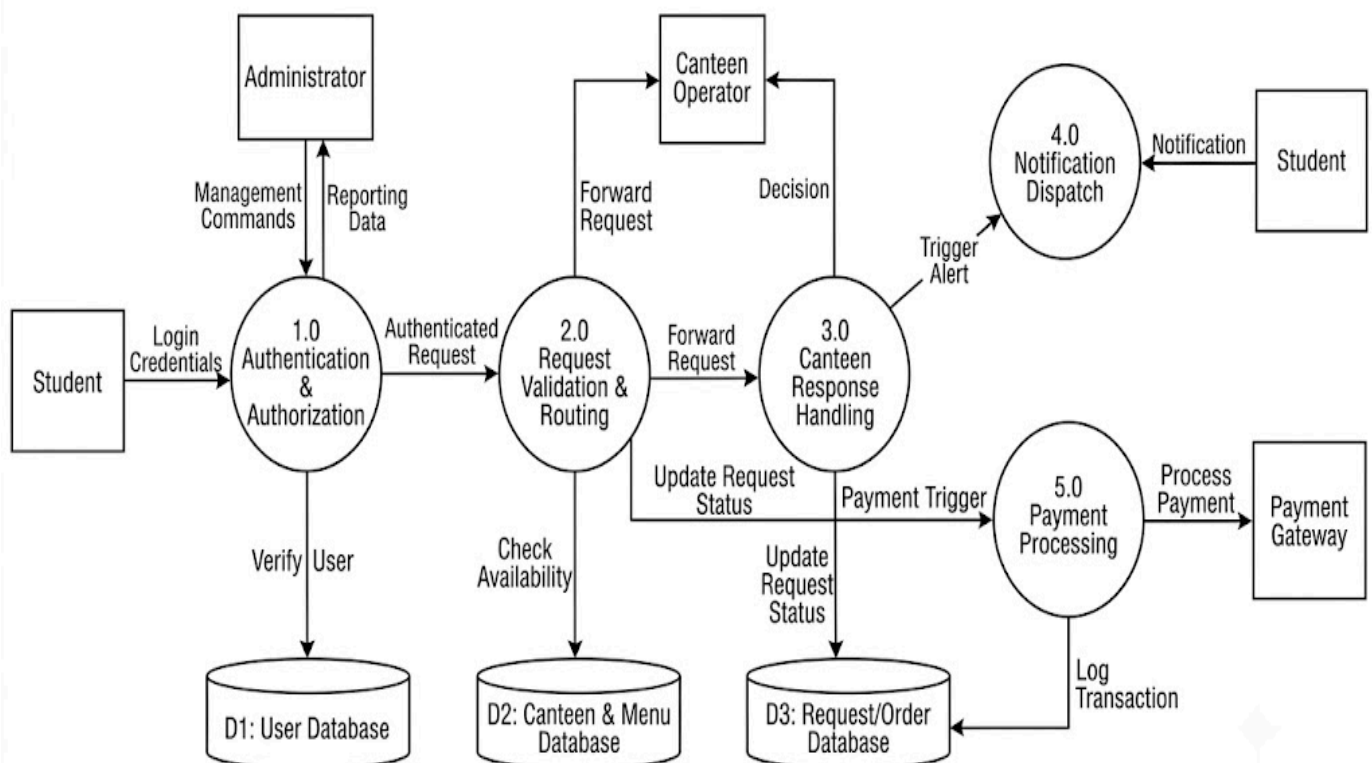


Fig 5.: Data Flow Level-1 (Decomposed Process View)

11. System Modules

Module	Key Functions
Student/Hostel Module	Login, menu browsing, request submission, status tracking, payment, feedback.
Faculty/Staff Module	Similar to Student/Hostel Module, primarily pickup-oriented.
Canteen Operator Module	Menu management, request acceptance/rejection, preparation time updates, item availability management.
Admin Module	Cross-canteen monitoring, analytics, user management, complaint handling.
Notification Engine	Real-time push notifications, SMS notifications, and in-app alerts throughout the request lifecycle.
Payment Module	In-app payment initiation, payment gateway integration, transaction logging.
Analytics Module	Request volume analysis, peak-hour analysis, canteen performance comparison, report generation.

12. Core Features

1. Multi-canteen and food-court request routing through a single application
2. Digital, real-time menu browsing per canteen
3. Request-based submission with server-mediated approval (accept/reject/modify)
4. Real-time estimated preparation time updates
5. Pickup or eligibility-verified hostel delivery options
6. QR-code based pickup verification
7. In-app payment processing
8. Request/order history
9. Feedback and ratings per completed request
10. Inventory/item availability indicators and temporary canteen closure status
11. Administrative dashboard with live monitoring and analytics

13. Feasibility Study

- 1. Technical Feasibility:** The system relies on well-established, widely supported technologies, including a standard client-server web/mobile architecture, relational or document databases, real-time notification services (such as push notification APIs), and standard payment gateway integrations. No component requires research-grade or experimental technology, making the system technically feasible to develop within an academic project timeline using existing frameworks and libraries.
- 2. Operational Feasibility:** Successful operation depends primarily on canteen operator adoption. Each canteen requires at least one operator-facing device and a willingness to shift from manual to digital request handling. Given the relatively small and controlled scope of a single campus, operator training and onboarding are manageable, and a pilot deployment with one canteen can be used to validate the workflow before wider implementation.
- 3. Economic Feasibility:** As an internally developed academic and institutional tool, the primary costs include development effort, hosting or server infrastructure, and ongoing maintenance. These costs are significantly lower than licensing a commercial food-delivery platform and eliminate the per-transaction commission fees typically charged by third-party delivery services. The proposed system is therefore economically feasible and justified by reduced recurring costs, improved operational efficiency, and better management of campus food services.

14. Technology Stack

The technology choices below reflect a standard, well-supported web and mobile development stack suited to a request-routing system with real-time notification and payment requirements. The selected technologies prioritize reliability, scalability, maintainability, and ease of implementation within an academic project timeline while avoiding experimental or niche tools.

Layer	Suggested Technology	Reason for Choice
Frontend (Student/Faculty/Staff App)	React.js (Web) or Flutter (Cross-Platform Mobile)	Component-based user interface, strong community support, and suitable for real-time request status updates.

Layer	Suggested Technology	Reason for Choice
Frontend (Canteen Operator Dashboard)	React.js	Provides a fast, form-based interface for request acceptance, rejection, menu management, and preparation time updates. Suitable for desktop or tablet usage.
Frontend (Admin Dashboard)	React.js with Chart.js or Recharts	Supports data-intensive dashboards and analytical visualizations through a mature charting ecosystem.
Backend / API Layer	Node.js with Express.js or Django (Python)	Handles authentication, business logic, request routing, and REST API development with strong library and real-time support.
Database	PostgreSQL or MongoDB	PostgreSQL is suitable for structured relational data, while MongoDB is appropriate for document-oriented data models.
Real-Time Notifications	Firebase Cloud Messaging (FCM) or Socket.IO	Enables instant notifications for request acceptance, rejection, preparation status, and pickup without continuous polling.
Authentication	JWT Authentication with optional College SSO Integration	Ensures secure login, verified institutional identity, and role-based access control.
Payment Gateway	Razorpay (or equivalent UPI-supported gateway)	Widely adopted in India, simple integration, and supports secure digital payments.
Hosting / Cloud	Institution Server, AWS, Render, or Railway	Suitable for pilot deployment with low infrastructure cost and future scalability.
Version Control	Git and GitHub	Industry-standard version control system supporting collaborative software development and project management.

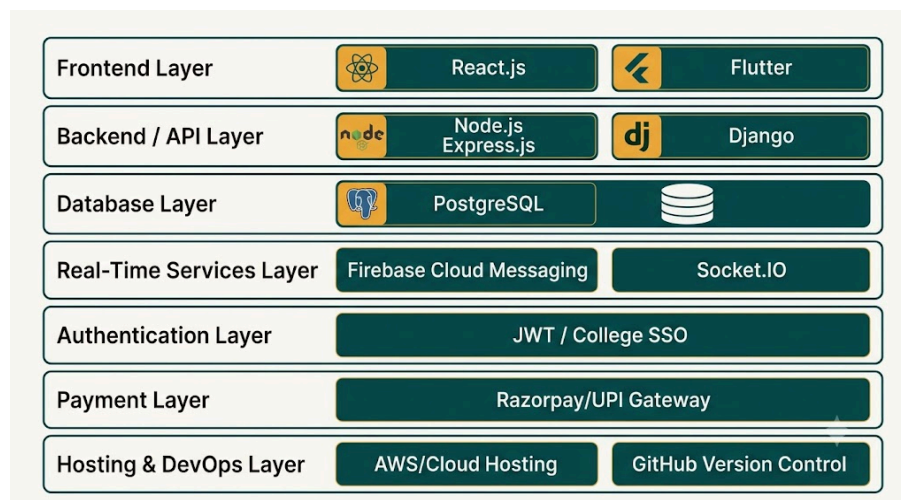


Fig 6.: Proposed Technology Stack

15. Future Scope

1. AI-based demand prediction to help canteens anticipate peak-hour rushes and reduce food wastage
2. Dynamic staffing recommendations for canteen operators based on predicted demand
3. Integration with a campus ID-linked digital wallet or canteen credit system
4. Extension to multi-campus deployment for institutions with more than one campus
5. Voice-based or chatbot-assisted request submission for accessibility
6. Sustainability tracking, including food waste analytics per canteen

16. Conclusion

The Smart Campus Canteen Management & Food Request Routing System addresses a clear and specific gap in campus food service delivery — one that generic commercial food-delivery platforms are not designed to solve, given their direct-order, open-market model. By introducing a centralized, request-routing architecture mediated by a college server, the system enables structured, transparent, and efficient food service across canteens, food courts, and hostel messes, while giving canteen operators practical tools for demand management and college administration comprehensive oversight through a unified monitoring dashboard. The system architecture, use case, and data flow diagrams presented in this report establish the technical foundation for the subsequent design and implementation phases of the project.